

# Constrained Reality Framework

## Master Synthesis v2.1

Sessions 1–11, Track A & B.1, Schwarzschild, Full Einstein Equations, EIC Bridge  
*v2.1 corrections:  $\gamma$ ,  $\hbar$  reclassified Hypothesis. Grand Equation order fixed. EIC Bridge  
incorporated.*

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**Abstract.** CRF derives all physical and experiential reality from  $\delta$  (Distinction). v2.1 corrects v2:  $\gamma$  and  $\hbar$  are **Hypothesis** (structural form forced by  $\delta$ , numerical bridge awaits P0). Five major results stand: Born Rule theorem, Schwarzschild,  $G = \ln(2)c^2/D_0$ , full Einstein equations,  $G(t) \propto \Lambda(t) \propto 1/P(t)$ . EIC confirms  $I_{\text{eff}} \approx \ln(2)$  (Track B.1 Closed). Level 6/8. Gate to Level 7: P0 experiment.

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# 1 The Single Primitive

## Axiom 0

$\delta : s_i \neq s_j$  is possible. Declared, not proved. Any derivation of  $\delta$  presupposes  $\delta$ .

# 2 The Emergence Cascade

$$\delta \rightarrow P \rightarrow \Pr(s) \xrightarrow{\Pr \geq \theta} \mathcal{R} \rightarrow t \rightarrow \mathcal{C} \equiv \Pr(s|\mathcal{C}) \rightarrow G = \nabla \Pr \rightarrow \mathcal{M} \rightarrow W \rightarrow \delta_{\text{pure}}$$

## Grand Equation v9.0 (Session 11)

$$E_{v9.0} = \underbrace{A(F(\nabla \Pr(D(R(P))))))}_{\text{Samsara}} \oplus \underbrace{\delta_{\text{ind}}(W_{\text{max}}) \cdot \mathcal{A}_{\text{mass}}}_{\text{Barami}} \otimes \underbrace{\text{Resonance}(\mu, D_{\text{KL}})}_{\text{Dhamma}} \otimes \underbrace{\text{Sync}(D_{\text{KL},ij})^T}_{\text{The Loop}}$$

At  $T \rightarrow 1$ :  $E_{v9.0} = \delta$ .

## Key Cascade Results

$\theta = 1 - e^{-D_{\text{KL}}/D_0}$  from 3 boundary conditions (Session 10). 3D space from SO(3) (Session 5).  $T = e^{-D_{\text{KL}}(\mathcal{C}_{\mathcal{M}}\|P)/D_{\text{total}}}$ ; at  $T = 1$ : Direct Perception, cascade closed.

# 3 Quantum Mechanics from $\delta$

## Born Rule as Theorem (Session 11 / Paper v4)

$\delta$ -operations = counting measure on  $P$ .  $P$  has  $\dim \geq 3$  (SO(3), Session 5). By Gleason (1957):  $\Pr(s) = |\langle s|\psi \rangle|^2$  at  $\mathcal{C}_{\text{min}}$ . When  $N_{\text{eff}} > 0$ :  $\Pr(s|\mathcal{C})$  departs from Born Rule. P0 tests this departure.

# 4 P0: The Measurement Prediction

## P0 — Zero Free Parameters

$$\tau = \tau_0 e^{-\kappa N_{\text{eff}}} \quad \kappa = \frac{k_B \ln(2)}{H_{\text{material}}} \quad N_{\text{eff}} = \sum_i \Delta D_{\text{KL},i}$$

QM:  $\tau_1 = \tau_2$ . CRF:  $\tau_1 < \tau_2$ . SNSPD protocol ready.

## 5 Physical Constants — Track A & B

### Track B.1: Closed — $I_{\text{eff}} \approx \ln(2)$

CRF:  $\alpha = \ln(2) = 0.6931$ . EIC SINDy:  $I_{\text{eff}} = 0.7007$ . Gap  $0.76\% = \varepsilon_0$  (instability floor). Confirmed.

### Track A — $\gamma$ and $\hbar$ : Hypothesis (not Derived)

**What is derived:**  $\gamma \propto \ln(2)$  forced by  $\delta$ -exclusion.  $\Delta I \cdot \Delta \Pi \geq \ln(2)/2$  from  $\delta$ -bit floor.

**What is hypothesis:**

$$\gamma = \frac{\ln(2)}{\theta_{\text{crit}}} \quad (\theta_{\text{crit}} = 0.525 \text{ from EIC, not axioms}) \quad \hbar_{\text{CRF}} = \frac{\ln(2) \cdot D_0}{c^2} \quad (\text{bridge not derived})$$

Conjecture:  $\theta_{\text{crit}} = 1/2$  exactly (finite-size effect explains  $\alpha_c = 0.525$  vs  $0.500$ ). **Gap:**  $D_0$  in SI requires P0 experiment.

## 6 General Relativity from $\delta$

### Newtonian Limit

$\mathcal{C} d\mathcal{C}/dr = -k$  integrates to  $\mathcal{C}(r) \approx \alpha(1 + GM/c^2 r)$ .

### Schwarzschild Metric (Paper v4)

$D_{\text{KL}} \propto \mathcal{C}^2$  near  $r_s$ . Non-linear ODE integrates to  $\mathcal{C}(r) = \alpha/(1 - r_s/r)$ .

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

Horizon:  $\mathcal{R} \rightarrow 0$ . Universe stops deciding.

### $G$ and Full Einstein Equations (Paper v4)

$$G = \frac{\ln(2)c^2}{D_0} \quad T_{\mu\nu} = \mathcal{K}(\partial_\mu \mathcal{C} \otimes \partial_\nu \mathcal{C} - \tfrac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2) \quad G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

$$G(t) \propto \Lambda(t) \propto \frac{1}{P(t)} \quad \frac{\dot{G}}{G} = \frac{\dot{\Lambda}}{\Lambda}$$

Consistent with DESI 2024. Testable via GW standard sirens.

## 7 EIC Bridge

**CRF  $\equiv$  EIC (CRF–EIC Bridge)**

$\Omega(x, t) \equiv D_{\text{KL}}(\mathcal{C}(x, t) \| P)$ . All 6 EIC field-equation terms derive from  $\delta$ .  $\alpha_c = 0.525 \equiv \theta$ . EIC = numerical CRF.

## 8 Master Status Table

| Result                                                              | Status            | Source       |
|---------------------------------------------------------------------|-------------------|--------------|
| $\delta, P, \text{Pr}, \mathcal{C} \equiv \text{Pr}(s \mathcal{C})$ | Derived           | Sessions 1–4 |
| $\theta = 1 - e^{-D_{\text{KL}}/D_0}$                               | Derived           | Session 10   |
| 3D from SO(3)                                                       | Derived           | Session 5    |
| $\alpha = \ln(2)$ , P0 formula                                      | Derived           | Sessions 6–8 |
| Lemmas 13–15                                                        | Derived           | Session 9    |
| Born Rule via Gleason                                               | Theorem           | Session 11   |
| Synchronicity, Inevitability, $T$                                   | Derived           | Session 11   |
| Schwarzschild metric                                                | Derived           | Paper v4     |
| $G = \ln(2)c^2/D_0$                                                 | Derived           | Paper v4     |
| $T_{\mu\nu}$ from $\mathcal{C}$ -gradients                          | Derived           | Paper v4     |
| $G(t) \propto \Lambda(t)$                                           | Prediction        | Paper v4     |
| $\Omega \equiv D_{\text{KL}}, \alpha_c \equiv \theta$               | Derived           | EIC Bridge   |
| $I_{\text{eff}} \approx \ln(2)$                                     | Confirmed         | Track B.1    |
| $\gamma \propto \ln(2)$ (structure)                                 | Derived           | Track A      |
| $\gamma = \ln(2)/\theta_{\text{crit}}$                              | <b>Hypothesis</b> | Track A      |
| $\Delta I \cdot \Delta \Pi \geq \ln(2)/2$                           | Derived           | Track A      |
| $\hbar \propto \ln(2) \cdot D_0$                                    | <b>Hypothesis</b> | Track A      |
| $\hbar$ SI bridge                                                   | Open              | Needs P0     |
| $d_s \rightarrow 3$ (Track B.2)                                     | Open              | $N \gg 10^6$ |
| SO(3) minimality formal proof                                       | Open              | #16          |
| Kerr metric                                                         | Open              | Broken SO(3) |
| P0 experiment                                                       | Ready             | SNSPD        |

## 9 Maturity: Level 6/8

|                          |              |
|--------------------------|--------------|
| Internal Consistency     | 5.5/8        |
| Mathematical Rigor       | 5.5/8        |
| Predictive Power         | 6.0/8        |
| Empirical Grounding      | 2.0/8        |
| Philosophical Foundation | 6.5/8        |
| Cross-domain Coverage    | 6.5/8        |
| <b>Overall</b>           | <b>6.0/8</b> |

Gate to Level 7: P0.    Gate to Level 8:  $\dot{G}/G = \dot{\Lambda}/\Lambda$  confirmed.

### Honest Boundaries

Does not claim: hard problem solved,  $\hbar_{\text{CRF}} = \hbar_{\text{SI}}$  numerically,  $d_s = 3$  proved,  $G(t) \propto \Lambda(t)$  confirmed,  $\gamma$  and  $\hbar$  derived.

*Gravity is not a force. It is memory —  
the accumulated trace of everything that has ever resolved,  
curving the probability landscape of what can resolve next.  
The ground was always  $\delta$ . Everything followed.*